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Lower Extremity Strains in the US National Football League, 2015-2019

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Background: Lower extremity (LEX) strains, including hamstring, quadriceps, adductor, and calf strains, are among the most common injuries in sports. These injuries lead to high burden, resulting in significant missed participation time.

Purpose: To describe the incidence of LEX strains in professional American football.

Study Design: Descriptive epidemiology study.

Methods: This study included all players who played in ≥1 National Football League (NFL) game or sustained a LEX strain during participation in the 2015-2019 seasons. LEX strain frequency was calculated by setting (game, practice, conditioning), timing in season (offseason, preseason, regular season, postseason), and roster position. Game incidence rates were calculated for season, roster position, and play type. LEX strains were identified in the standardized leaguewide electronic health record (n = 32 teams).

Results: Across 5 years, 5780 LEX strains were reported among 2769 players (1-year risk, 26.7%; 95% CI, 26.0%-27.3%); 69% (n = 4015) resulted in time loss. Among all LEX strains, 54.7% were hamstring (n = 3163), 24.1% adductor (n = 1393), 12.6% calf (n = 728), 8.3% quadriceps (n = 477), and 0.3% multiple muscle groups (n = 19). Most were reported during preseason practices (n = 1076; 27%) and regular season games (n = 1060; 26%). The 2-week period of training camp practices comprised 19% of all time-loss strains. Among game injuries, preseason games had the highest rate of LEX strain (2.9/10,000 player-plays; 95% CI, 2.6-3.2). Defensive secondary players accounted for the highest proportion of time-loss LEX strains (27%; n = 1082). In games, punt plays had nearly twice the injury rate of kickoff plays (14.9/1000 plays [95% CI, 13.1-17.0] vs 7.5/1000 plays [95% CI, 6.2-8.9], respectively) and >3 times the rate of pass plays (4.3/1000 plays; 95% CI, 4.0-4.7) and run plays (2.6/1000 plays; 95% CI, 2.3-2.9). In aggregate, LEX strains led to an estimated 16,748 participation days missed each year and a median 12 days missed per injury.

Conclusion: LEX strains affected 1 in 4 NFL players each year, resulting in a high burden of injury in terms of time lost from practice and competition. Safe return to the NFL season during training camp and reduction of injuries during regular season games are key focuses for future injury reduction.

Keywords: lower extremity strain; hamstring strain; adductor strain; quadriceps strain; NFL; football; epidemiology

Injuries to the lower extremity (LEX)—specifically, those to the hip, knee, and ankle—affect a high number of National Football League (NFL) football players annually and can lead to considerable missed time from practice and game participation.[7] A recent study utilizing the NFL's electronic health record (EHR) database identified that LEX strains—namely, those to the hamstring, adductor, quadriceps, and calf muscle groups—accounted for a burden of injury (i.e., a metric that considers the number of injuries and associated time lost from participation) that approached that of the top burden injury: anterior cruciate ligament (ACL) injury (10,251 vs 12,369 player-days missed per season owing to preseason, regular season, and postseason injuries).[7] A previous study examining injuries from preseason training camp through the end of the preseason on 1 NFL team over a 10-year period reported that muscle strains were the most common type of injury in preseason practices, with hamstring strains being the most prevalent.[7] In a similar study, hip muscle strains accounted for 60% of all hip injuries in the NFL from 1997 to 2006. Although not as prevalent as hamstring and quadriceps strains, the incidence of calf strains is approximately 2 per team per year.[6,11]

While some information has been published on LEX strains in the NFL,[3-6,11] most of these previous studies have been limited to data from a single NFL team or data derived from publicly available sources. The purpose of this study was to investigate and describe the incidence of LEX strains sustained by NFL players by using the leaguewide EHR database. This study is intended to be a foundational epidemiologic study of incidence of LEX strains in the NFL from 2015 to 2019. It can be used to drive future research and innovation related to injury reduction initiatives, such as strength and conditioning strategies as

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players return to the NFL season, preseason acclimation and schedules, improvements to postinjury rehabilitation procedures and return-to-play decision making for secondary prevention, and other prevention programs.

METHODS

We conducted a descriptive epidemiology study of LEX strains, defined as hamstring, adductor, quadriceps, and calf strains, in the NFL during the 2015-2019 years (including offseason, preseason, regular season, and postseason). All players who played in ≥1 NFL game or who sustained a LEX strain during NFL participation (game or nongame) were included. Approvals were obtained through the NFL Player Scientific and Medical Research Protocol[8] and by the Mt Sinai Institutional Review Board (Pro00060188).

LEX Strain Inclusion Criteria

All LEX strains (primary and recurrent) diagnosed by team medical staff and entered into the leaguewide EHR were included, regardless of whether the injury caused the player to miss time from football activity. LEX strains that occurred outside the team environment or were reported as not football related were excluded from the analysis. Information about the specific muscle groups involved was obtained from standardized clinical impression codes within the injury diagnosis.

Data Sources

Data were obtained from the leaguewide EHR used by all 32 NFL teams continuously and in a standardized manner throughout the study period.[2] In addition to recording information about the LEX strain diagnosis, team medical staff recorded data on timing of the injury within the year, activity at the time of the injury, and return to participation within the EHR. The EHR undergoes thorough data quality reviews, including regular training on standardized injury reporting guidelines, auditing against media reports to ensure completeness of data entry, and weekly data reviews to flag inconsistencies or irregularities in the data.[2]

NFL EHR data were linked with the Game Statistics and Information System, which provides the player's roster position at the time of the injury, years of NFL service, and the number of plays that teams and/or individual players participated in during games. Participation information was not available for practices, and subsequently, incidence rate calculations focus on game injuries and game participation.

Definitions

LEX strain incidence was described overall and stratified by player characteristics and injury characteristics, such as whether the LEX strain resulted in time loss, the timing of the injury within the NFL year, and the activity at the time of the injury. Time-loss injuries were defined as those that resulted in removal from participation and inability to return to play or take full part in future football activities for a period of time after the injury, as determined by team medical staff. Timing of the injury within the NFL year included offseason, preseason, regular season, and postseason. Offseason begins at the start of the club's offseason program and finishes at the end of the club's mandatory minicamp or with organized team activities. Preseason begins at the veteran report date and finishes after the club's final preseason game. Preseason injuries were stratified by those that occurred during training camp, defined as the first 2 weeks of practices before the first preseason game beginning on the veteran report date. The timing of game-incurred LEX strains was described by week, timing in season (preseason, regular season, postseason), and quarter in game. Activity was reported in several ways, including the team activity at the time of the injury (game, practice, conditioning, other/unknown), the play type during which the injury occurred for game injuries (punt, kickoff, pass, run, field goal, extra point), and the practice activity during which the injury occurred for practice injuries (11-on-11, 9-on-9, individual, etc). Player characteristics included the number of years of service (0 indicates the rookie year) and roster position. A roster position of "special teams" consisted of the kicker, punter, punt returner (not including punt returners who play another position), or long snapper. For injuries during special teams plays, position at the time of injury was categorized for kickoffs as kick return unit, kicking unit, kick returner, or kicker. For punts, position at time of injury was categorized as punt return unit, punt unit, long snapper, punter, or punt returner.

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TABLE 1
Distribution of Lower Extremity Strains in the National Football League, 2015-2019

	2015		2016		2017		2018		2019		5-y Mean	
	No.	%	No.	%	No.	%	No.	%	No.	%	No.	%
Total	1122	100	1095	100	1168	100	1229	100	1166	100	1156	100
Time-loss status												
Time-loss	700	62.4	782	71.4	820	70.2	855	69.6	858	73.6	803	69.5
Non-time-loss	422	37.6	313	28.6	348	29.8	374	30.4	308	26.4	353	30.5
Season												
Offseason	272	24.2	281	25.7	283	24.2	313	25.5	264	22.6	283	24.4
Preseason	372	33.2	354	32.3	402	34.4	431	35.1	424	36.4	397	34.3
Regular season	463	41.3	451	41.2	469	40.2	473	38.5	455	39.0	462	40.0
Postseason	15	1.3	9	0.8	14	1.2	12	1.0	23	2.0	15	1.3
Team activity												
Game	406	36.2	369	33.7	439	37.6	398	32.4	392	33.6	401	34.7
Practice	596	53.1	603	55.1	577	49.4	676	55.0	647	55.5	620	53.6
Conditioning	96	8.6	103	9.4	127	10.9	113	9.2	88	7.5	105	9.1
Other/unknown	24	2.1	20	1.8	25	2.1	42	3.4	39	3.3	30	2.6
Strain type[a]												
Hamstring	643	57.3	632	57.7	647	55.4	637	51.8	604	51.8	633	54.7
Adductor	229	20.4	216	19.7	279	23.9	353	28.7	316	27.1	279	24.1
Calf	157	14.0	155	14.2	131	11.2	148	12.0	137	11.7	146	12.6
Quadriceps	92	8.2	90	8.2	105	9.0	88	7.2	102	8.7	95	8.3
Multiple muscles	1	0.1	2	0.2	6	0.5	3	0.2	7	0.6	4	0.3

[a]The number of injuries for multiple groups were adductor and hamstring (n = 10), adductor and quadriceps (n = 5), hamstring and calf (n = 3), and hamstring and quadriceps strain (n = 1).

Days Missed and Injury Burden

The number of days missed was reported by team medical staff and calculated as the number of days between the date that the player was removed from participation owing to injury and the date of return to full participation (not inclusive). A standardized definition for return to full participation was provided to team medical staff, which was defined as the date that the player was able to return to full participation without restriction, regardless of team activity or timing in the year. Subsequently, the number of days missed was recorded continuously throughout the full year and not specific to return to game participation.

Injury burden per year, which combines the number of injuries with the number of days missed from participation, was calculated by the mean annual number of days missed for LEX strains. For injuries where the number of days missed was not available (eg, owing to a player leaving the team), the number of days missed was estimated as the 2015-2019 median days missed for the same type of injury.

Statistical Analysis

Descriptive statistics were calculated for all variables of interest and presented as count (percentage), mean (standard deviation), and/or median (range and interquartile range [IQR]), where appropriate. The 1-year risk of LEX strain was calculated as the number of players who sustained a LEX strain each year (2015-2019) divided by the total population at risk for injury in that specific year, with the population defined by unique player signings per year, regardless of duration that the player participated in the NFL within the year (2015, n = 3278; 2016, n = 3288; 2017, n = 3337; 2018, n = 3333; 2019, n = 3167). The exact binomial distribution was used to calculate the 1-year risk of LEX strain and accompanying 95% CIs.

Incidence rates were calculated using several playing time metrics: number of plays (including 22 players participating in 1 play) and number of player-plays (1 player participating in 1 play). For injury rates on kickoff and punt plays, it was assumed that each play was composed as follows: 10 players on the kick return unit, 10 players on the kicking unit, 1 kick returner, and 1 kicker; 10 players on the punt return unit, 9 players on the punt unit, 1 long snapper, 1 punter, and 1 punt returner. Poisson regression was used to calculate incidence rates and accompanying 95% CIs.

RESULTS

Overall LEX Strain Incidence

Across 5 years, 5780 LEX strains were reported among 2769 players, 69.5% (n = 4015) of which resulted in time loss (Table 1). The annual mean number of LEX strains was 1156 per year across the 5-year study period. The overall 1-year risk of sustaining a LEX strain was 26.7% (95% CI, 26.0%-27.3%). Among all LEX strains, 54.7% were hamstring (n = 3163), 24.1% adductor (n = 1393), 12.6% calf (n = 728), 8.3% quadriceps (n = 477), and 0.3% multiple muscle groups (n = 19). LEX strains were most frequent during the regular season (n = 2311; 40.0%) and preseason

TABLE 2
Distribution of Time-Loss Lower Extremity Strains in the National Football League, 2015-2019

	2015		2016		2017		2018		2019		5-y Mean	
	No.	%	No.	%	No.	%	No.	%	No.	%	No.	%
Total	700	100	782	100	820	100	855	100	858	100	803	100
Season												
Offseason	99	14.1	212	27.1	203	24.8	238	27.8	216	25.2	194	24.1
Preseason	279	39.9	259	33.1	317	38.7	308	36.0	318	37.1	296	36.9
Regular season	313	44.7	304	38.9	294	35.9	302	35.3	316	36.8	306	38.1
Postseason	9	1.3	7	0.9	6	0.7	7	0.8	8	0.9	7	0.9
Team activity												
Game	287	41.0	273	34.9	303	37.0	264	30.9	288	33.6	283	35.2
Practice	370	52.9	432	55.2	429	52.3	496	58.0	496	57.8	445	55.4
Conditioning	36	5.1	70	9.0	80	9.8	75	8.8	62	7.2	65	8.0
Other/unknown	7	1.0	7	0.9	8	1.0	20	2.3	12	1.4	11	1.3
Years of service												
0 (rookie)	121	17.3	177	22.6	161	19.6	176	20.6	172	20.0	161	20.1
1	120	17.1	134	17.1	139	17.0	137	16.0	152	17.7	136	17.0
2	90	12.9	108	13.8	120	14.6	149	17.4	120	14.0	117	14.6
3	86	12.3	88	11.3	83	10.1	95	11.1	99	11.5	90	11.2
4	61	8.7	62	7.9	86	10.5	82	9.6	81	9.4	74	9.3
5	69	9.9	62	7.9	76	9.3	60	7.0	69	8.0	67	8.4
6	40	5.7	52	6.6	45	5.5	36	4.2	47	5.5	44	5.5
≥7	113	16.1	99	12.7	110	13.4	120	14.0	118	13.8	112	13.9
Roster position												
Offensive line	36	5.1	74	9.5	62	7.6	89	10.4	75	8.7	67	8.4
Running back	68	9.7	75	9.6	82	10.0	69	8.1	84	9.8	76	9.4
Tight end	35	5.0	49	6.3	50	6.1	56	6.5	66	7.7	51	6.4
Quarterback	7	1.0	7	0.9	3	0.4	0	0.0	8	0.9	5	0.6
Wide receiver	117	16.7	150	19.2	169	20.6	168	19.6	171	19.9	155	19.3
Defensive secondary	207	29.6	211	27.0	220	26.8	232	27.1	212	24.7	216	26.9
Defensive line	108	15.4	71	9.1	81	9.9	93	10.9	102	11.9	91	11.3
Linebacker	109	15.6	129	16.5	141	17.2	132	15.4	129	15.0	128	15.9
Special teams	13	1.9	16	2.0	12	1.5	16	1.9	11	1.3	14	1.7

(n = 1983; 34.3%) and during practice (n = 3099; 53.6%) or games (n = 2004; 34.7%).

Time-Loss LEX Strain Incidence

Regarding the 4015 time-loss LEX strains, the incidence remained the highest during the regular season (n = 1529; 38.1%) and preseason (n = 1481; 36.9%) and during practice (n = 2223; 55.4%) or games (n = 1415; 35.2%) (Table 2). Most occurred during preseason practices (n = 1076; 27%) and regular season games (n = 1060; 26%). Defensive secondary (n = 1082; 26.9%), wide receiver (n = 775; 19.3%), and linebacker (n = 640; 15.9%) players sustained the most time-loss LEX strains over the 5-year study period.

Time-Loss LEX Strains in Games

Among game injuries, the time-loss injury rate was 2.4 LEX strains per 10,000 player-plays (95% CI, 2.2-2.5), with the highest rate during preseason games (2.9/10,000 player-plays; 95% CI, 2.6-3.2) (Table 3). During regular season games, time-loss LEX strains occurred most frequently during weeks 2 to 4—on average, 17 to 20 time-loss LEX strains are reported annually in games each week during weeks 2 to 4—and then decreased gradually throughout the regular season starting in week 5 (Figure 1). Additionally, regular season time-loss game LEX strains occurred more commonly in the second and fourth quarters (n = 273 and 265, respectively) as compared with the first and third (n = 180 and 218). While defensive secondary players, linebackers, and wide receivers accounted for nearly two-thirds (64.3%) of injuries in games, punters and kickers had the highest time-loss injury rate (3.8/10,000 player-plays; 95% CI, 2.7-5.3), followed by running backs (3.3/10,000 player-plays; 95% CI, 2.7-3.9).

In terms of the play type at the time of the injury during games, time-loss LEX strains were more likely on special teams plays, particularly on punt plays, than on run and pass plays. Although pass plays resulted in the highest number of time-loss LEX strains (n = 537; 38.0%), punt plays resulted in 232 injuries (16.4%) and had nearly twice the rate of kickoff plays (14.9/1000 plays [95% CI, 13.1-17.0] vs 7.5/1000 plays [95% CI, 6.2-8.9], respectively) and >3 times the rate of pass (4.3/1000 plays; 95% CI,

TABLE 3
Time-Loss Lower Extremity Strain Rates in National Football League Games, 2015-2019

	No. of Time-Loss Game Injuries	No. of Player-Plays	Injury Rate per 10,000 Player-Plays (95% CI)
Overall	1415	6,018,765	2.4 (2.2-2.5)
Season			
Preseason	335	1,161,463	2.9 (2.6-3.2)
Regular season	1060	4,657,221	2.3 (2.1-2.4)
Postseason	20	200,081	1.0 (0.6-1.5)
Roster position[a]			
Offensive line	124	1,224,298	1.0 (0.8-1.2)
Running back	113	344,469	3.3 (2.7-3.9)
Tight end	59	360,059	1.6 (1.3-2.1)
Quarterback	17	229,910	0.7 (0.5-1.2)
Wide receiver	190	661,839	2.9 (2.5-3.3)
Defensive secondary	440	1,368,232	3.2 (2.9-3.5)
Defensive line	158	832,030	1.9 (1.6-2.2)
Linebacker	280	907,508	3.1 (2.7-3.5)
Punter/kicker	34	90,413	3.8 (2.7-5.3)
		No. of Plays	Injury Rate per 1000 Plays (95% CI)
Play type[b]			
Punt	232	15,522	14.9 (13.1-17.0)
Kickoff	123	16,506	7.5 (6.2-8.9)
Pass	537	123,897	4.3 (4.0-4.7)
Run	226	87,895	2.6 (2.3-2.9)
Field goal	11	6326	1.7 (1.0-3.1)
Extra point	5	7402	0.7 (0.3-1.6)

[a]n = 7 player-plays could not be categorized with respect to roster position.
[b]n = 281 game injuries occurred during other or unknown play types.

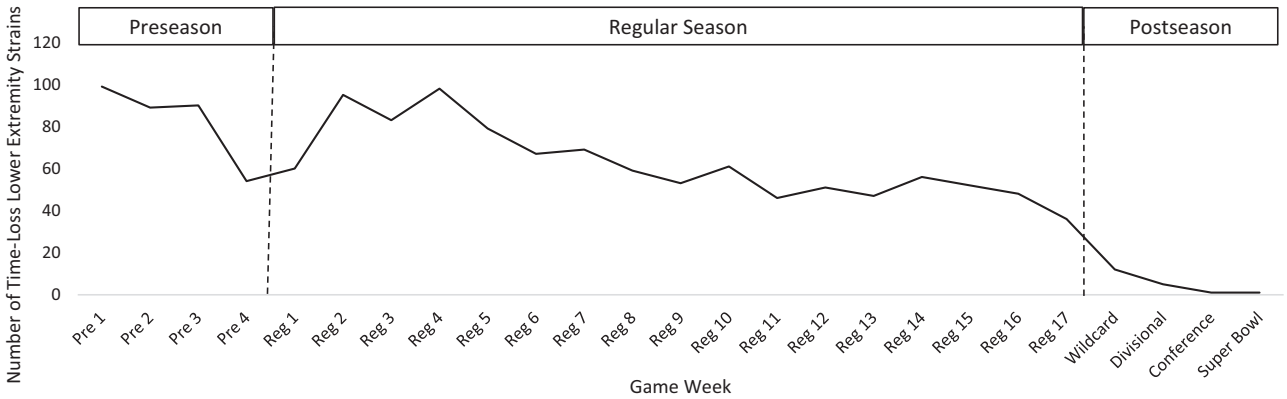

Figure 1. Time-loss lower extremity strains during games by week in the National Football League, 2015-2019. Three time-loss lower extremity strains reported during the Hall of Fame game and 1 during the Pro Bowl are not presented in this figure. Pre, preseason; Reg, regular season.

4.0-4.7) and run (2.6/1000 plays; 95% CI, 2.3-2.9) plays (Table 3). Among the time-loss LEX strains during punt plays, the punt unit on the coverage team sustained the greatest number (n = 162; 69.8%) and highest rate (11.6/10,000 player-plays; 95% CI, 9.9-13.5) of LEX strains (Figure 2). On kickoff plays, the kickoff unit on the coverage team sustained the greatest number (n = 85; 69.1%) of LEX strains, but kickers had the highest rate (9.7/10,000 player-plays; 95% CI, 5.9-15.8). Excluding punters and kickers, the roster positions with the highest proportion of game injuries resulting from a punt or kickoff play were linebackers (38.6%; 108/280), defensive secondary players (31.1%; 137/440), running backs (30.1%; 34/113), and wide receivers (18.4%; 35/190).

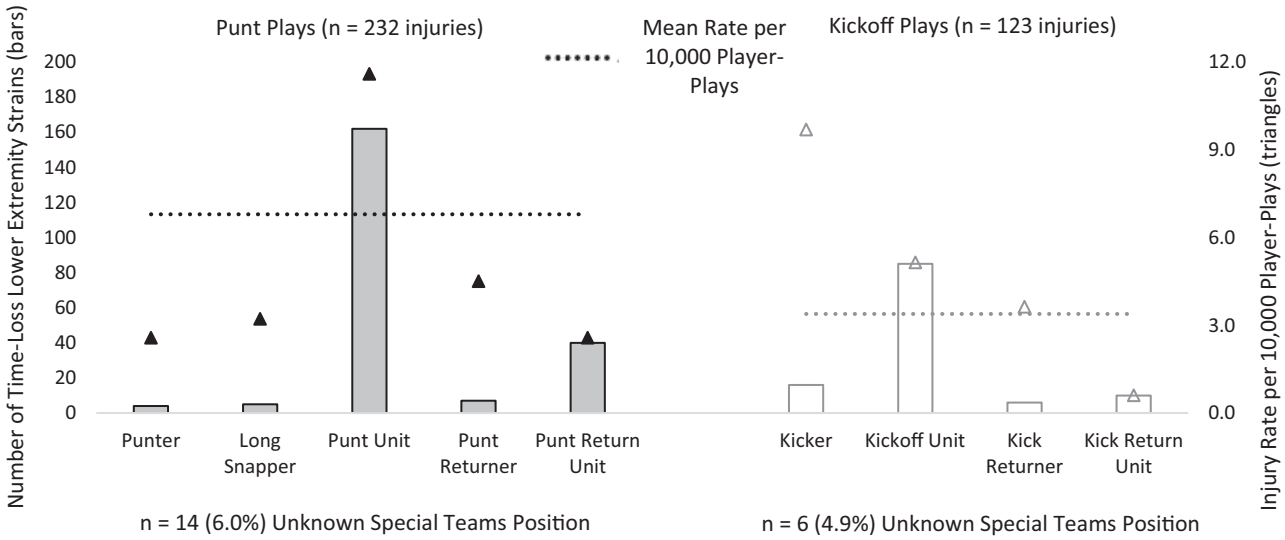

Figure 2. Time-loss lower extremity strains on punt and kickoff plays by position in National Football League games, 2015-2019.

TABLE 4
Distribution of Time-Loss Lower Extremity Strains by Season in National Football League Practices, 2015-2019[a]

	Offseason		Preseason		Regular Season	
	No.	%	No.	%	No.	%
Total	703	100	1076	100	433	100
Years of service						
0 (rookie)	227	32.3	253	23.5	84	19.4
1	151	21.5	181	16.8	83	19.2
2	108	15.4	176	16.4	52	12.0
3	61	8.7	115	10.7	50	11.5
4	51	7.3	108	10.0	31	7.2
5	39	5.5	74	6.9	34	7.9
6	20	2.8	57	5.3	24	5.5
≥7	46	6.5	112	10.4	75	17.3
Team activity						
11-on-11	301	42.8	464	43.1	203	46.9
Individual	153	21.8	99	9.2	51	11.8
Special teams	48	6.8	141	13.1	60	13.9
1-on-1	24	3.4	110	10.2	29	6.7
7-on-7	46	6.5	81	7.5	33	7.6
Warm-up or stretching	7	1.0	12	1.1	13	3.0
Conditioning	17	2.4	7	0.7	2	0.5
9-on-7	3	0.4	9	0.8	4	0.9
9-on-9	2	0.3	2	0.2	0	0
Unknown	102	14.5	151	14.0	38	8.8

[a]Eleven postseason time-loss practice injuries are excluded from this table. Percentages are based on column total.

Time-Loss LEX Strains in Practices

LEX strains occurring in practice represented 55.4% (n = 2223) of all time-loss LEX strains. Of those, 1076 (48.4%) occurred during preseason and 703 (31.6%) during offseason (Table 4). A smaller proportion happened during practice in the regular season (n = 433; 19.5%) and postseason (n = 11; 0.5%). Across all years, practice LEX strains were most frequently reported during 11-on-11 activities (42.8%-46.9%). In the offseason, individual activities (n = 153; 21.8%) accounted for the next-highest frequency of time-loss LEX strains in practice, while special teams activities in preseason and regular season practices (n = 141 and 60; 13.1% and 13.9%, respectively) accounted for the next-highest frequency of injuries. Among time-loss LEX strains in preseason practice, 72% (n = 775) were

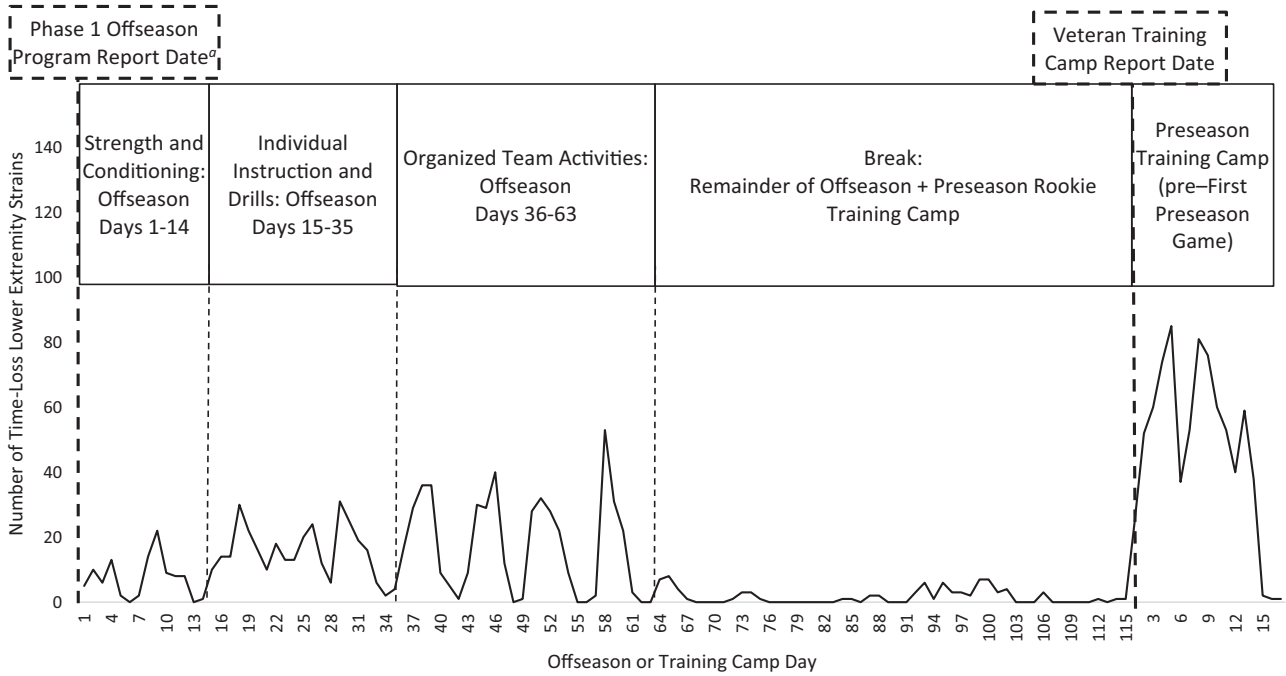

Figure 3. Time-loss lower extremity strains during offseason and training camp in the National Football League by day, 2015-2019.

reported during training camp. The training camp period saw the highest per-day number of time-loss LEX strains across the preseason, with a mean 5 to 17 time-loss LEX strains per day during this period (Figure 3). In particular, the fifth day after the training camp veteran report date—representing the period shortly after the league's 3-day contact acclimation period, when full contact practices begin—had the highest per-day number of time-loss LEX strains (n = 17 time-loss LEX strains each year, on average).

Days Missed and Injury Burden

Approximately 70% of LEX strains resulted in time loss. Among those, the median number of days missed from participation was 12 (IQR, 5-25) and the mean was 22.0 (SD, 36.9) (Table 5). Players with multiple muscle groups involved had the highest median number of days missed (17 days; IQR, 9-29), while players with an adductor strain had the lowest (7 days; IQR, 3-16). Median days missed were similar for calf strains, hamstring strains, and quadriceps strains (13-14 days). In aggregate, LEX strains led to an estimated 16,748 player-days missed from participation each year, driven largely by the estimated burden of hamstring strains, which contributed to an estimated 9648 player-days missed from participation each year.

DISCUSSION

Across our 5-year study, LEX strains affected approximately 1 in 4 NFL players each year (1-year risk, 26.7%; 95% CI, 26.0%-27.3%). Nearly 70% of LEX strains led to time loss—specifically, a median 12 days (IQR, 5-25). In aggregate, LEX strains led to an estimated 16,748 days missed from participation by players each year. This represents the highest burden injury grouping in the NFL when compared with other LEX injuries, topping the 12,369 days missed from participation owing to ACL tears each preseason, regular season, and postseason.[7] The primary reason for the high burden of LEX strains is the frequency at which they occur. While LEX strains result in players missing a median 12 days per injury as compared with 290 days for an ACL tear, approximately 803 time-loss LEX strains occurred each year in the NFL as compared with approximately 45 ACL tears per preseason, regular season, and postseason.[7] Over half of all LEX strains were hamstring strains (54.7%), and approximately one-quarter were adductor strains (24.1%), with calf (12.6%) and quadriceps (8.3%) strains making up a smaller proportion. These findings are similar to previously published studies that analyzed data from single teams,[3-6,11] but this study adds a more comprehensive assessment of LEX strains across the NFL population and the full year by leveraging the leaguewide EHR database. Given the relatively high incidence and burden, these injuries are a key focus for NFL injury reduction initiatives, and findings from this study suggest that preseason practices and regular season games are important focuses for reduction strategies, since 53% (n = 2136) of all time-loss LEX strains occur in these settings.

Preseason Practices

Among the 4015 time-loss LEX strains from 2015 to 2019, 27% occurred during preseason practices, including 775

TABLE 5
Days Missed due to LEX Strains in the National Football League, 2015-2019[a]

	No. of LEX Strains	No. of Time-Loss LEX Strains	Days Missed From Participation Owing to Time-Loss LEX Strains						LEX Strains Excluded From Days Missed[c]
			Estimated Mean Burden per Year[b]	Median	IQR	Mean	SD	Range	
Total	5780	4015	16,748	12	5-25	22.0	36.9	0-528	472
LEX strain type[d]									
Hamstring	3163	2347	9648	14	6-27	21.5	29.6	0-434	298
Adductor	1393	855	3128	7	3-16	19.3	43.8	0-528	73
Calf	728	474	2152	14	5-31	23.7	33.6	0-356	47
Quadriceps	477	323	1803	13	5-27	30.8	61.0	0-399	53
Multiple muscles	19	16	94	17	9-29	30.1	43.4	2-173	1

[a]IQR, interquartile range; LEX, lower extremity.
[b]Injury burden per year was calculated by summing days missed attributed to all LEX strains, then dividing by 5 (the number of years in this study). If an injury had unavailable return information owing to departure from the National Football League or missing data, the number of days missed was imputed by using the median days missed across 2015 to 2019 for the injury group of interest.
[c]Time-loss LEX strains excluded from days-missed calculation: the number of days missed could not be calculated for injury records where the player left the club before return to play, the player was still out from injury, or the record had an incomplete return date.
[d]The number of injuries for multiple groups were adductor and hamstring (n = 10), adductor and quadriceps (n = 5), hamstring and calf (n = 3), and hamstring and quadriceps (n = 1).

during the 2-week period of training camp practices, making up 19% of all time-loss LEX strains. The number of LEX strains during this relatively short period of the year—one that constitutes <10% of the active 28-week NFL season (from training camp through Super Bowl)—suggests that this time frame should be a key focus for future injury reduction initiatives.

In particular, the fifth day after the training camp veteran report date had the highest per-day number of time-loss LEX strains, and per-day injury counts remained high through the first 10 days of training camp. In the NFL during this study period, there was a 9-week noncontact offseason program that spanned mid-April to mid-June, followed by an approximately 1-month break before players returned for the preseason. During this study period, most players reported to preseason training camp on the veteran report date, and preseason training camp began with 3 days of contact acclimation before full-contact padded practices started. The fifth day after the training camp veteran report date was often the first or second day that players were exposed to full contact practice, and the spike in LEX strains during that time suggests that players' muscles may not have been fully acclimated to the demands of NFL practice intensity when full-contact practices begin. Understanding how best to acclimate to NFL participation after the return from offseason is a key focus for ongoing research to reduce LEX strains in the preseason, and more research is needed to understand how to maximize the benefits of building up training load to reduce injury risk.

Regular Season Games

Similar to preseason practices, where 27% of all time-loss LEX strains occurred during our study period, regular season games saw approximately one-quarter of LEX strains in the NFL (n = 1060; 26%), making this setting another key target of injury reduction initiatives.

Findings from our investigation into LEX strains during regular season games suggest similarities and differences in terms of injury reduction opportunities for these injuries as compared with those during preseason practices. Similar to preseason, we noted that the highest incidence of LEX strains was during the early regular season, particularly weeks 2 to 4. This suggests that appropriate acclimation to the demands of regular season game participation may be as important as that to the NFL preseason to reduce the risk of LEX strain early in the regular season. The decreasing trend of LEX strain incidence also suggests that fatigue related to chronic workload over a longer period of the season may not be a key factor influencing LEX strain occurrence; rather, acute introduction of new workload demands without appropriate acclimation may be a more important factor to consider for reduction of LEX strains. By quarter of game, LEX strains were more frequent in the second and fourth quarters than the first and third. One hypothesis is that fatigue from game participation may contribute to this pattern. Specifically, players may become more fatigued through the second quarter of the game. After the second quarter during halftime, players have some time off from playing to hydrate and recover. Additionally, football games may be more likely to be played at a higher tempo toward the end of the second and fourth quarters because of end-of-half and end-of-game situations, respectively, potentially leading to more injuries. Additional research is needed to understand optimal preparation for regular season game participation, at the individual player level and at the team level.

Within games, a key finding of this study was that, despite making up approximately 5% of all plays, LEX strains on punt plays represented about 16% of all time-loss LEX strains in games, leading to nearly twice the injury rate of kickoff plays and >3 times the rate of pass

and run plays. Among players participating on punt plays, most LEX strains occurred to the punt unit on the coverage team, and the punt unit had the highest rate of injury among punt-specific position groups. According to roster position, punters (in conjunction with kickers) also had a high rate of injury per player-play. These findings highlight a need for injury reduction strategies targeted toward players participating on punt plays, especially those on the punt unit and punters.

Furthermore, nearly two-thirds of time-loss game injuries were sustained by defensive secondary players, linebackers, and wide receivers. While many of the injuries to these players occur during typical run and pass plays, a substantial proportion of injuries to these players happened during punts and kickoffs (defensive secondary players, 31.1%; linebackers, 38.6%; wide receivers, 18.4%). Defensive secondary players and wide receivers frequently participate in punts and kickoffs as gunners (a player whose primary objective is to move "downfield as fast as possible to eventually tackle the punt returner"[10]) and returners.

Special teams plays, including punts and kickoffs, are unique within the NFL setting in that they often require a combination of greater speed, longer distance, quicker accelerations and decelerations, and more frequent adaptation to changing play dynamics than a typical run or pass play, which is particularly true for players on the punt unit (i.e., coverage team players). Given these requirements, specific training recommendations may be beneficial for special teams players. In addition, there is often more time during the game between special teams plays than between consecutive run and pass plays. Given the increase in time between plays, extra in-game warm-up or preparation requirements may be needed for players to maintain readiness for the sprinting, decelerating, and changing-of-direction activities required during special teams participation over the course of the game. The unique aspects of these play types previously led to focused initiatives directed toward kickoff rule changes,[9] and the difference in LEX strain incidence rates on kickoffs versus punts in this study may suggest a need for reviewing the punt play structure and rules to identify potential opportunities for injury reduction. Additional research on these topics is needed to identify the most actionable potential strategies for injury reduction.

Clinical Management and Return to Play

Finally, while this study was primarily focused on understanding the incidence of LEX strains in this setting to drive future research and innovation related to injury reduction initiatives, the burden associated with LEX strains warrants additional investigation into factors influencing time loss given that approximately 70% of LEX strains resulted in days missed, with a median 12 days missed from participation. LEX strains are recognized to have a high rate of recurrence,[1] necessitating a focus on return-to-play pathways, treatments, and modifiable factors influencing recurrence. While a deep dive on secondary prevention and recurrence was outside the scope of this study, understanding the risk of LEX strain recurrence and identifying opportunities for secondary prevention in this population are key priorities, and research in this area is ongoing.

Limitations

Participation information for players (minutes/plays of participation) was not available for practices at the same level of granularity as games; thus, practice LEX strain rates could not be estimated. Information on the LEX strain grade was not consistently available during this study period; however, during annual EHR injury reporting training before the 2020 season, emphasis was placed on the importance of documenting LEX strain grade, especially for hamstring strains, bringing the level of missingness ("unknown grade") from 82% in 2015 to 16% in 2020 (unpublished data). Future research will leverage this improvement to better understand correlates and predictors of LEX strain grade and effects of LEX strain grade on injury burden. Injury burden metrics were calculated by the number of days missed attributed to the injury, regardless of the timing of injury in the NFL year. However, days missed may be influenced by factors not fully reflected in burden calculations, including football-related factors, such as players moving between teams and roster cuts, as well as other medical factors. Limitations of the days-missed metric also include that it does not capture the burden players bear related to pain management, and injuries with missing information on number of days missed may be more likely to miss more time, possibly leading to an underestimate of the true number of days missed.[7]

CONCLUSION

LEX strains affected 1 in 4 NFL players each year, resulting in a high burden of injury in terms of time lost from practice and competition. Safe return to the NFL season during training camp and reduction of injuries during regular season games are key focuses for future injury reduction.

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